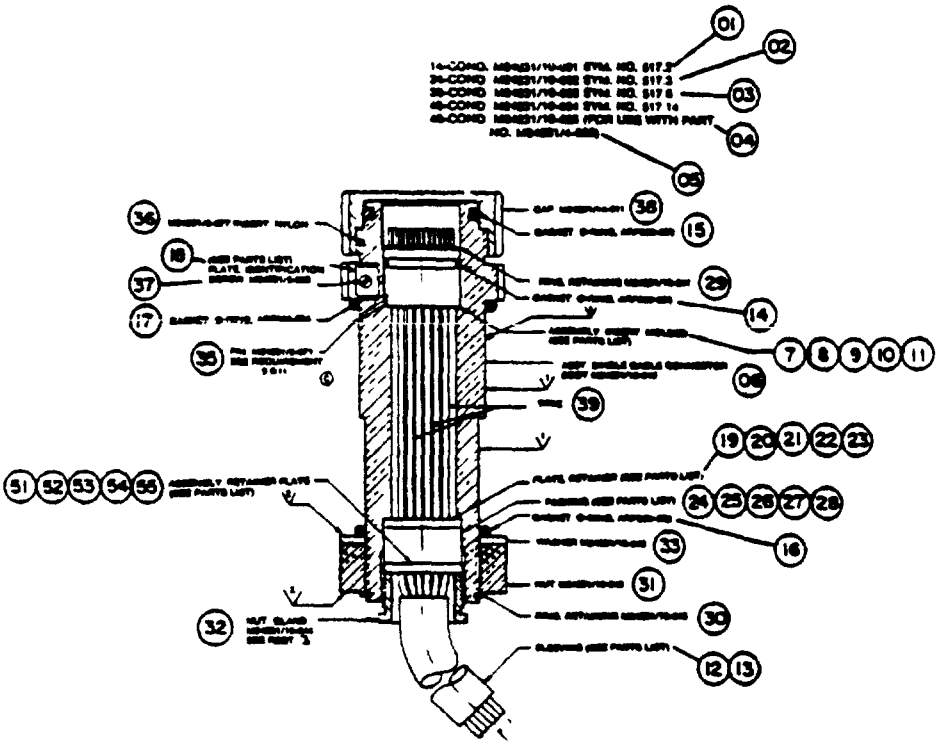
MILITARY SPECIFICATION SHEET

CONNECTORS, SINGLE CABLE, TYPE III, 14-, 24-, 30-, OR 40-CONDUCTOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the single cable connector described herein shall consist of this specification and the latest issue of MIL-C-24231.



1/ Hardness reading to be taken at these points.

2/ Hardness reading to be taken at this face at 120-degree intervals.

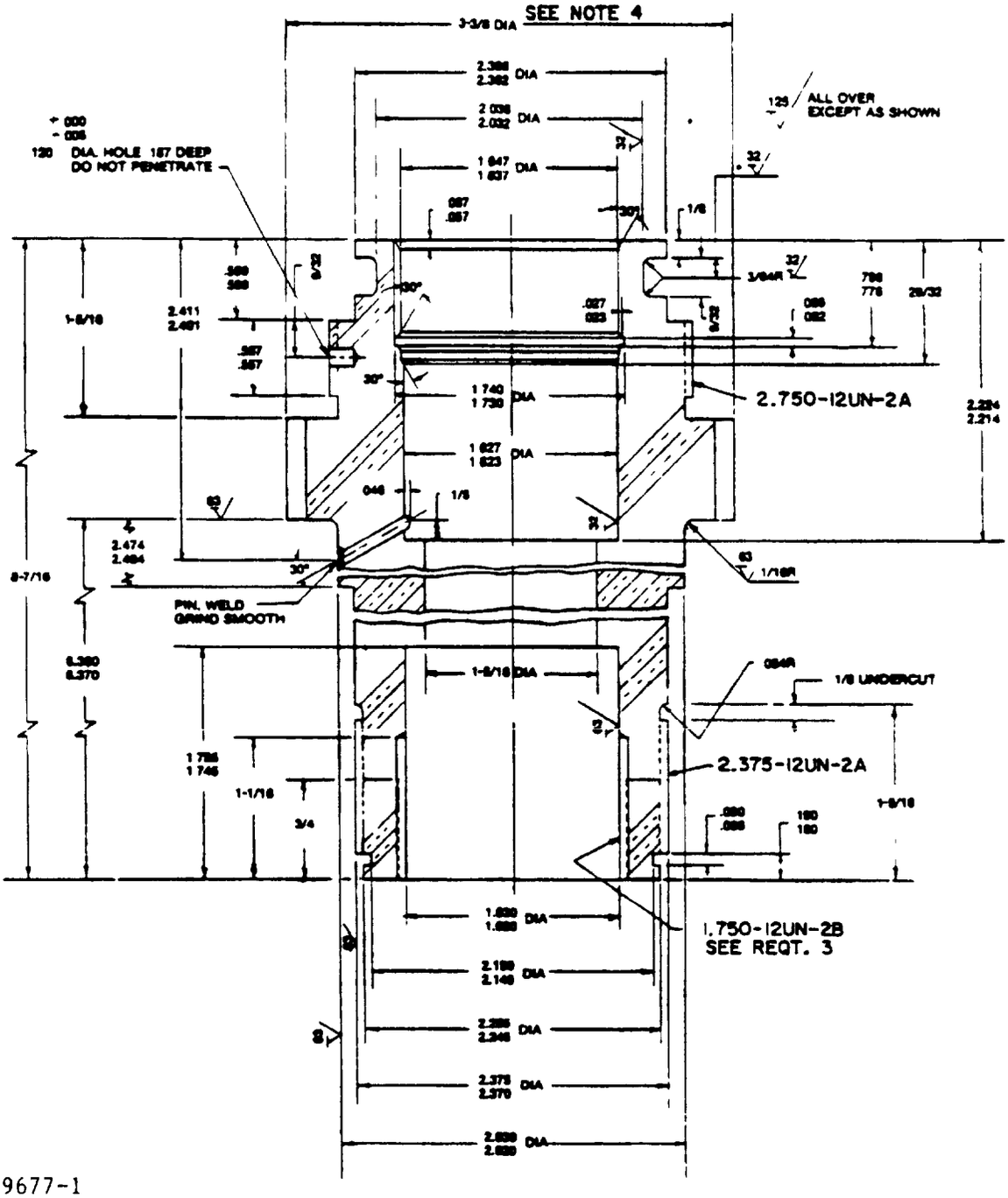
SH 9677

FIGURE 1. Connector, single cable, 14-, 24-, 30-, or 40-conductor.

AMSC N/A FSC 5935

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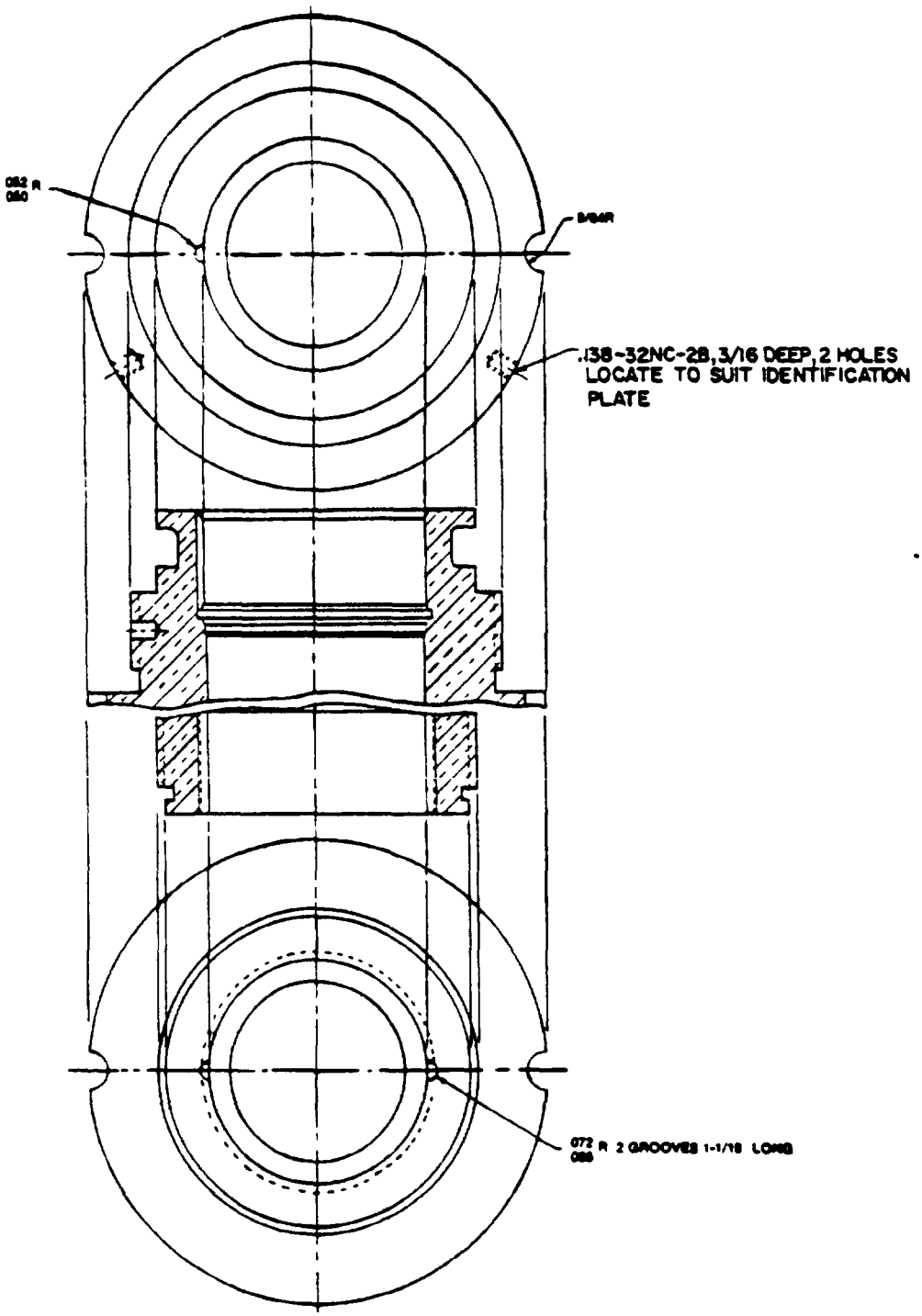
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SH 9677-1

FIGURE 2. Assembly, single cable connector body - M24231/10-040. (06)

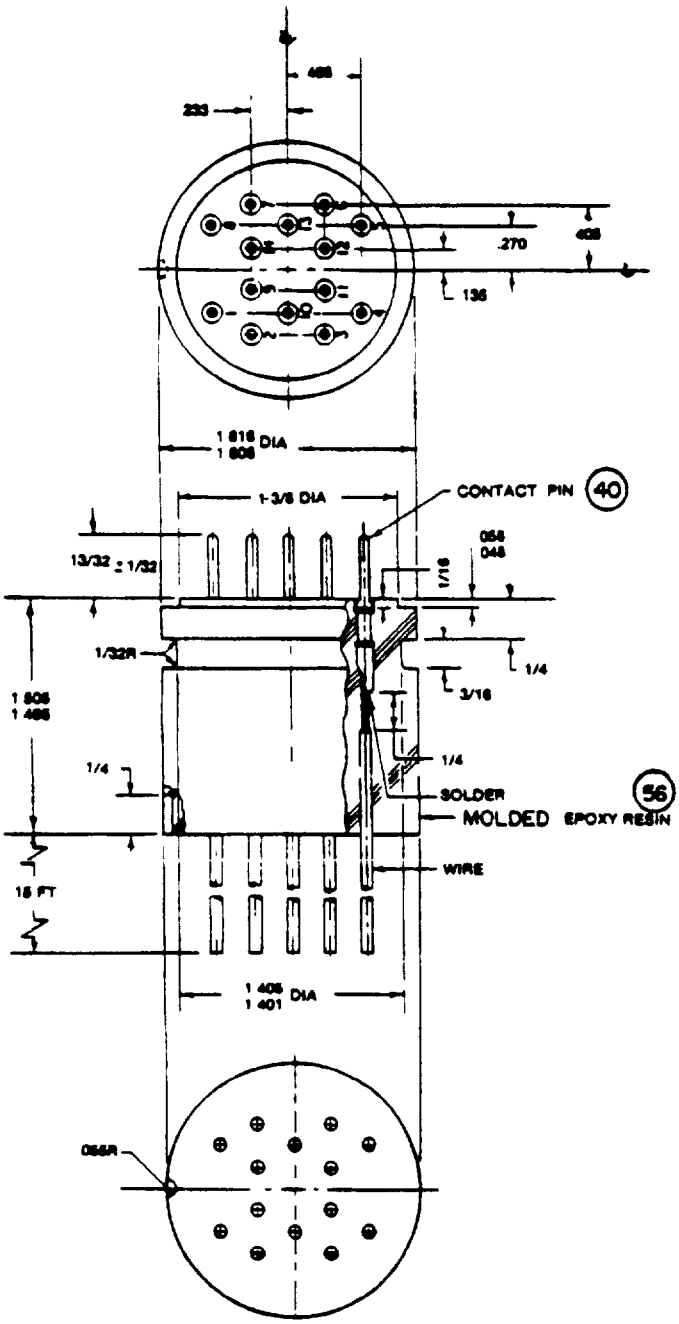
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SH 9677-2

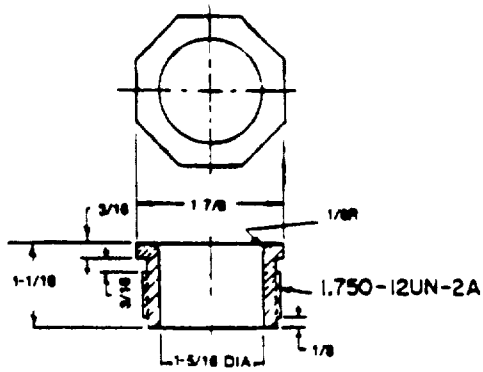
FIGURE 2. Single cable connector body - M24231/10-040. - Continued (06)

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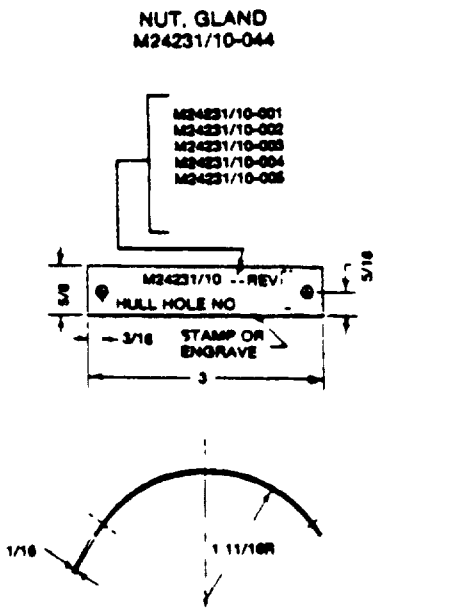
07

FIGURE 3. Assembly, insert, molded (14-conductor) - M24231/10-010.



32

FIGURE 4. Nut, gland - M24231/10-044.



18

FIGURE 5. Plate, identification.

SH 9677-3

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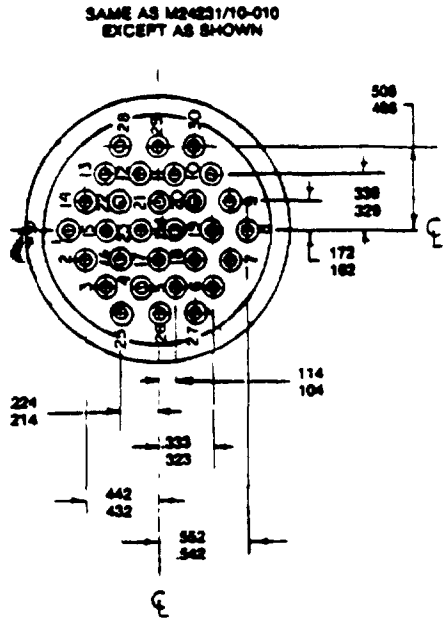


FIGURE 6. Assembly, insert, molded (30-conductor) - M24231/10-030. (09)

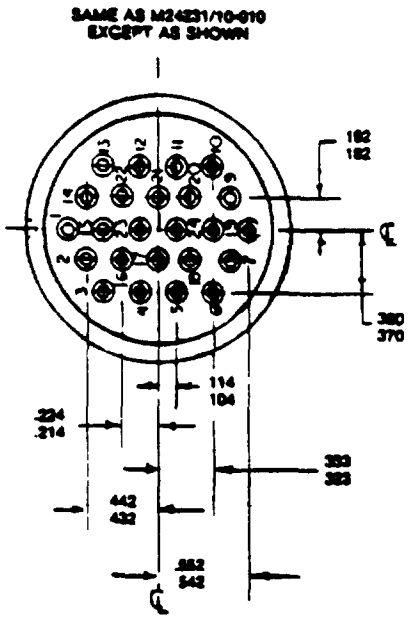


FIGURE 8. Assembly, insert, molded (24-conductor) - M24231/10-020. (08)

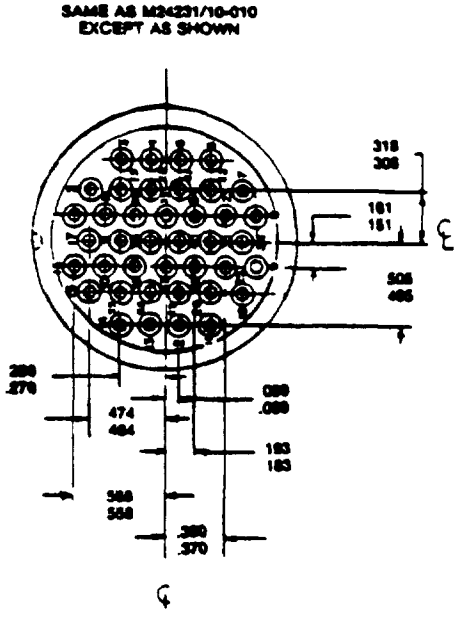


FIGURE 7. Assembly, insert molded (40-conductor) - M24231/10-035. (10)

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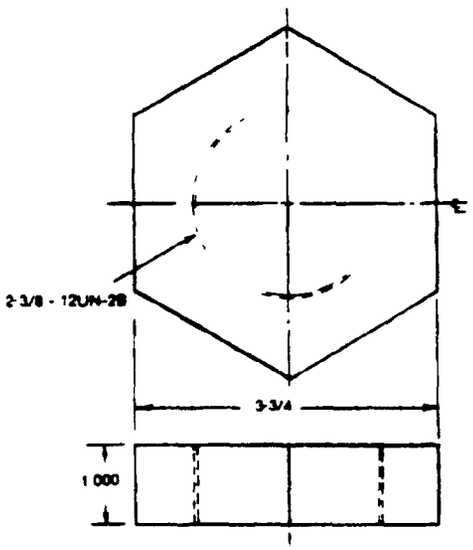


FIGURE 9. Nut - M24231/10-042. (31)

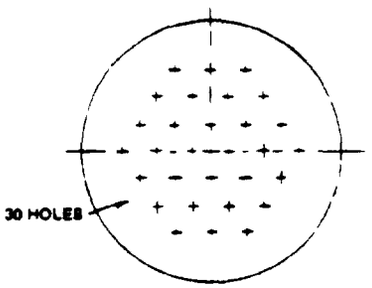


FIGURE 11. Plate, retainer (same as M24231/10-060 except no positioning pins required) - M24231/10-061. (21)

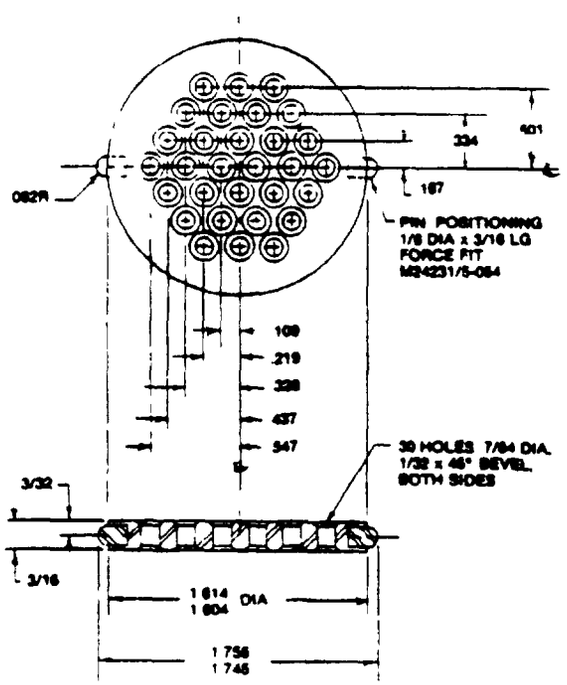
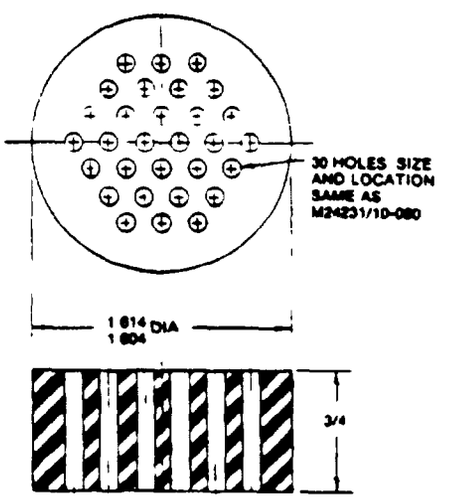


FIGURE 10. Assembly, plate retainer - M24231/10-060. (53)



(27)

FIGURE 12. Packing - M24231/10-062.

SH 9677-8

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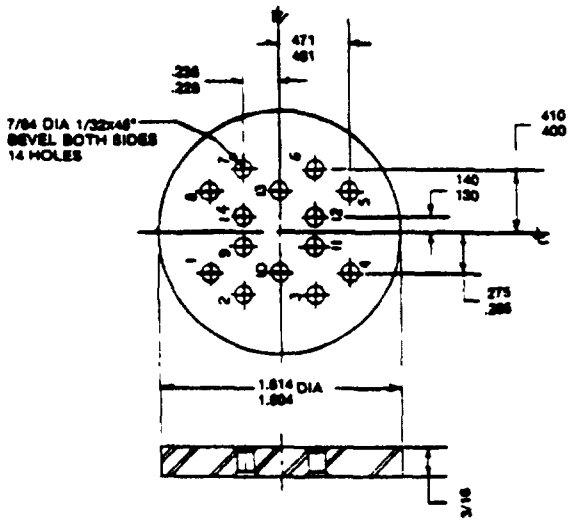


FIGURE 13. Plate, retainer - M24231/10-061. (19)

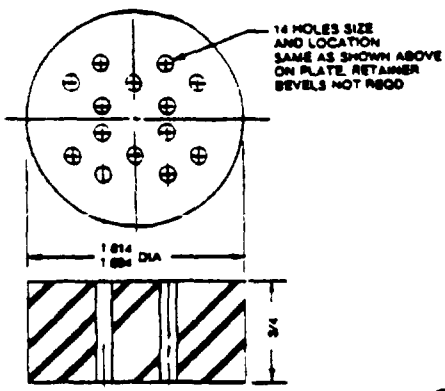


FIGURE 14. Packing - M24231/10-062. (24)

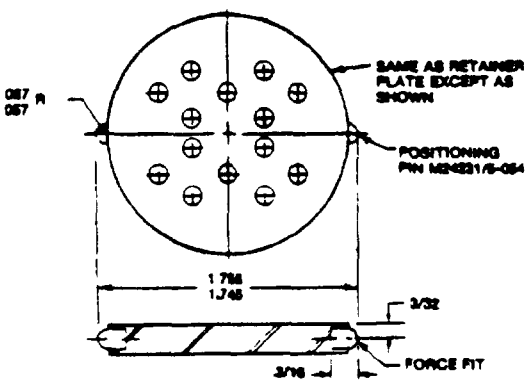


FIGURE 15. Assembly, retainer plate - M24231/10-060. (51)

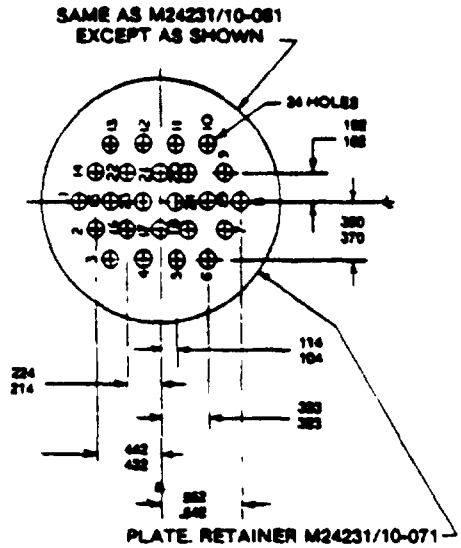
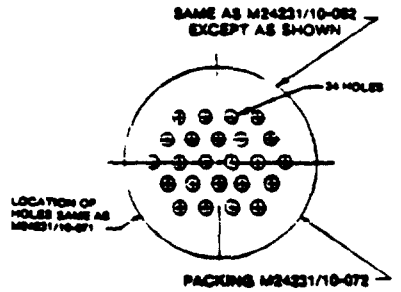
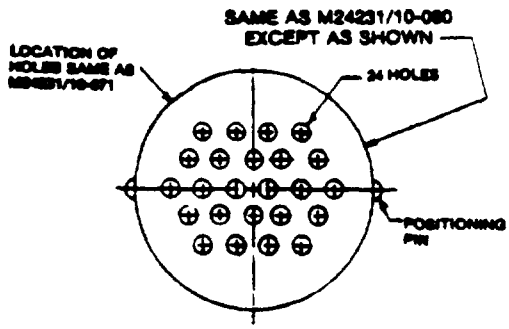


FIGURE 16. Plate, retainer - M24231/10-071. (20)



(25)

FIGURE 17. Packing - M24231/10-072.



(52)

FIGURE 18. Assembly, retainer plate - M24231/10-070.

SH 9677-9

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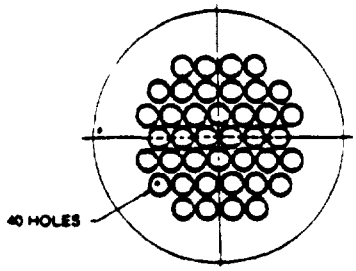


FIGURE 20. Plate, retainer (same as M24231/10-077 except no positioning pins required) - M24231/10-078. (22)

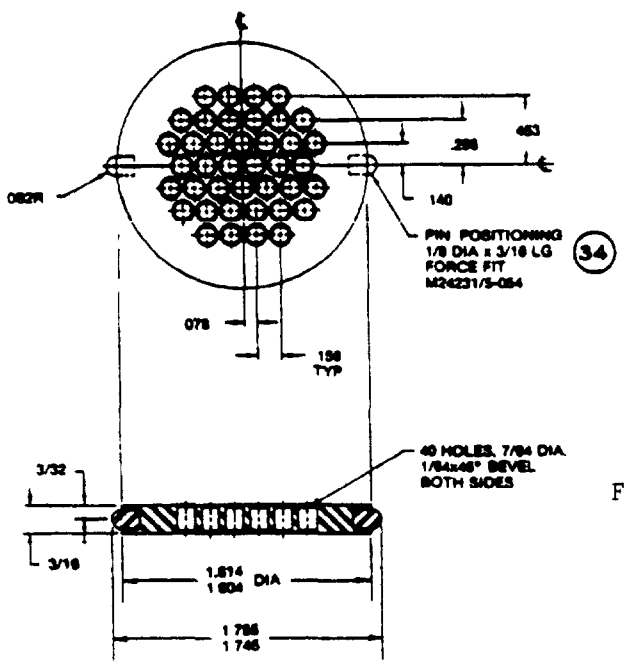


FIGURE 19. Assembly plate retainer - M24231/10-077. (54)

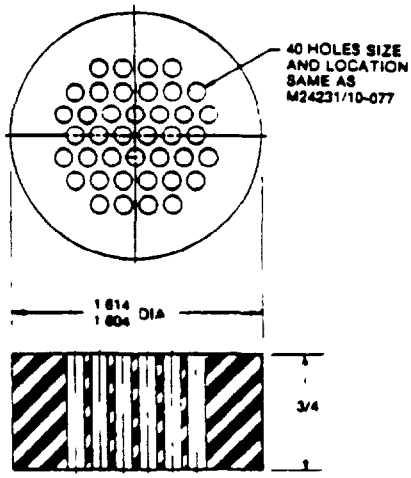


FIGURE 21. Packing - M24231/10-079. (27)

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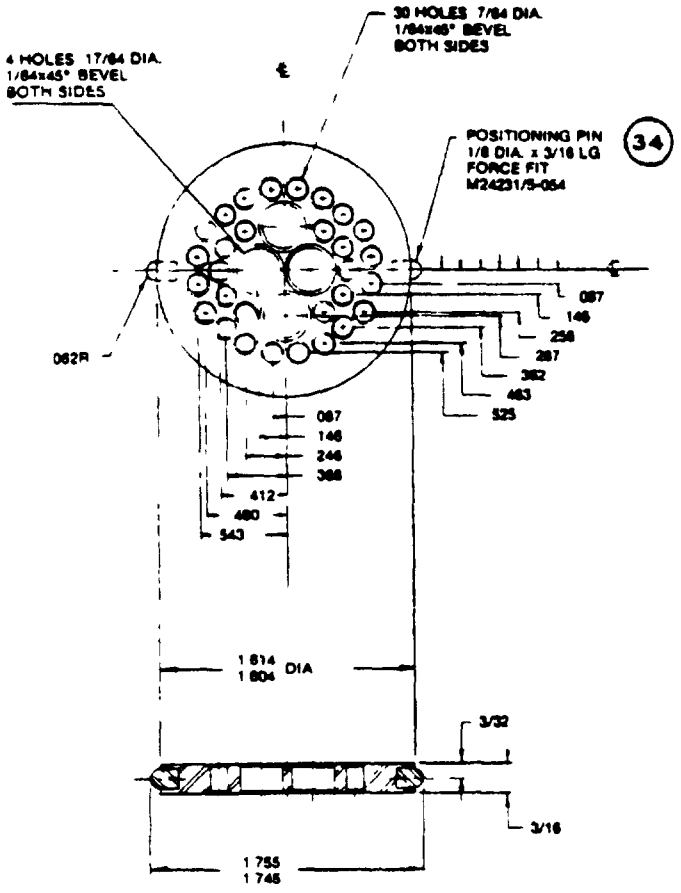


FIGURE 22. Assembly, plate, retainer - M24231/10-074.

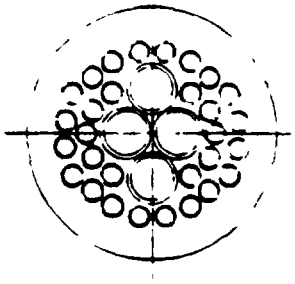


FIGURE 23. Plate, retainer (same as M24231/10-074 except no positioning pins required) - M24231/10-075. (23)

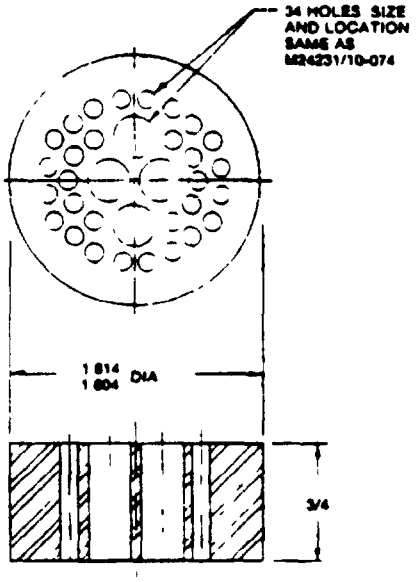
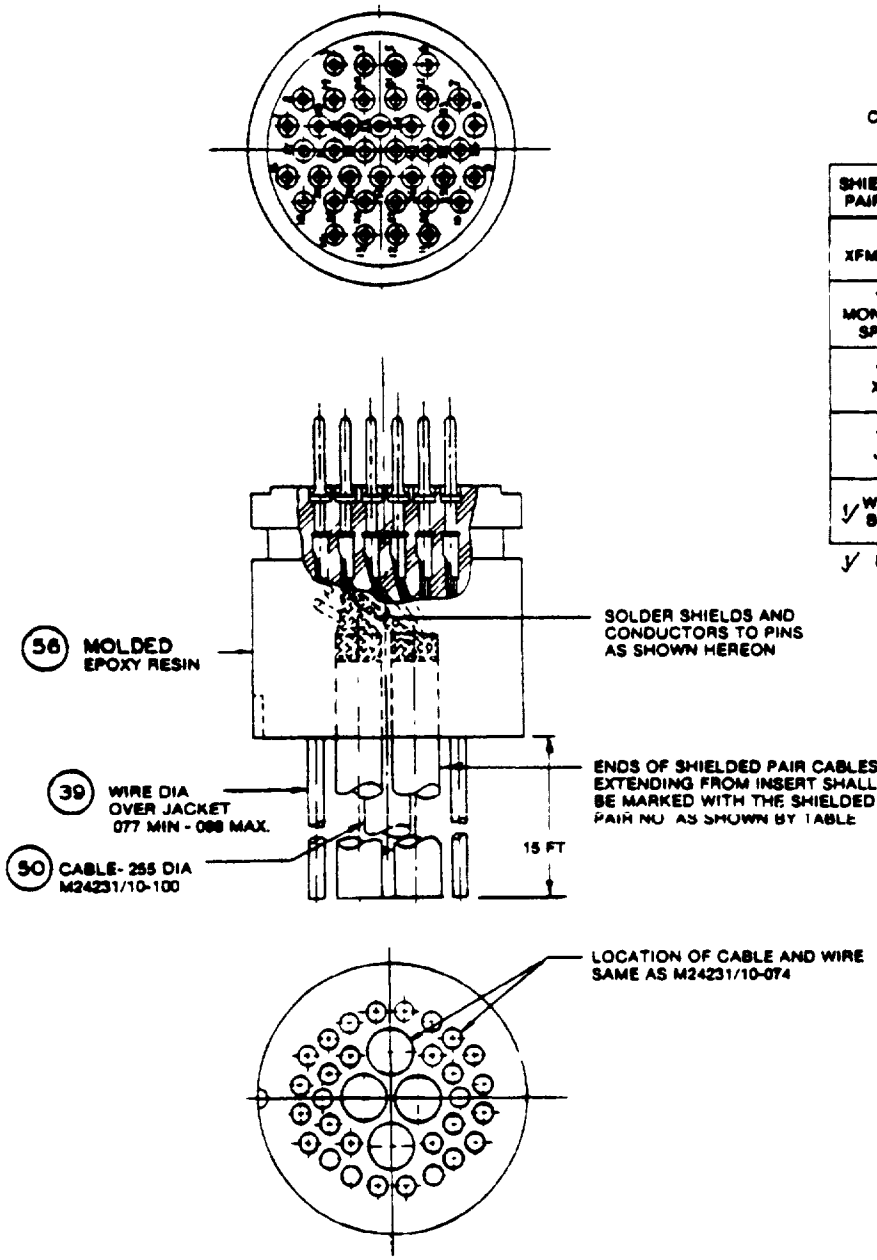


FIGURE 24. Packing - M24231/10-076. (28)

SH 9677-11

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CONNECTIONS FOR SHIELDED PAIR-2SWU-1

SHIELDED PAIR NO	COLOR	PIN NO
15 XFMR SW	WHITE BLACK SHIELD	30 29 31
16 MONITOR SPKR	WHITE BLACK SHIELD	33 32 31
18 XU	WHITE BLACK SHIELD	37 36 36
19 JA	WHITE BLACK SHIELD	40 39 36
1/ WIRE 8-16	BLACK	ALL OTHER PINS

1/ MIL-W-16878/1

SH 9677-12

FIGURE 25. Assembly, insert, molded (40-conductor) - M24231/10-036 (same as M24231/10-035 except as shown). (11)

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REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. The gland nut shall be tightened until packing tightens against the wires and cables. This can be determined by moving wires or cables gently back and forth until wire or cable is seized by packing. The gland shall be turned approximately one-fourth of a turn more to hold the wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.
3. Full or partial threads shall not extend more than 0.083 inch beyond the 3/4-inch thread length.
4. Refer to requirement 4 of MIL-C-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.
5. The retainer plates and retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with class A, form 1, hot finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with class A, form 1, cold drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.
6. See requirement 7 of MIL-C-24231/5.

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TABLE I. Parts list.

Item 1/ no.	Description 2/	Quantity for M24231/10- 3/					Material/specification	Remarks
		001	002	003	004	005		
01	Connector assembly, single cable, 14-conductor, symbol no. 517.2	1					-----	See figure 1
02	Connector assembly, single cable, 24-conductor, symbol no. 517.3		1				-----	See figure 1
03	Connector assembly, single cable, 30-conductor, symbol no. 517.6			1			-----	See figure 1
04	Connector assembly, single cable, 40-conductor, symbol no. 517.14				1		-----	See figure 1
05	Connector assembly, single cable, 40-conductor					1	-----	See figure 1
06	Connector body, single cable	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 2 and requirement 4
07	Insert assembly, molded, 14-conductor	1					See items 39, 40, and 56	See figure 3
08	Insert assembly, molded, 24-conductor		1				See items 39, 40, and 56	See figure 3
09	Insert assembly, molded, 30-conductor			1			See items 39, 40, and 56	See figure 3
10	Insert assembly, molded, 40-conductor				1		See items 39, 40, and 56	See figure 3

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/10- 3/ 001	002	003	004	005	Material/specification	Remarks
11	Insert assembly, molded, 40-conductor					1	See items 39, 40, and 56	See figure 3
12	Sleeving, 0.680 id by 0.040 thick, 15 ft	1					Type F, form U, category 1, color clear, MIL-I-631	See figure 1
13	Sleeving, 0.890 id by 0.050 thick, 15 ft		1	1	1	1	See item 12	See figure 1
14	O-ring, gasket	1	1	1	1	1	MS28775-220	See figure 1
15	O-ring, gasket	1	1	1	1	1	MS28775-329	See figure 1
16	O-ring, gasket	1	1	1	1	1	MS28775-332	See figure 1
17	O-ring, gasket	1	1	1	1	1	MS28775-334	See figure 1
18	Plate, identification	1	1	1	1	1	CRES or Al, MIL-P-15024	See figure 5
19	Plate, retainer	1					Ni-Cu, class A, QQ-N-281	See figure 13 and requirement 5
20	Plate, retainer		1				See item 19	See figure 16 and requirement 5
21	Plate, retainer			1			See item 19	See figure 11 and requirement 5
22	Plate, retainer				1		See item 19	See figure 20 and requirement 5
23	Plate, retainer					1	See item 19	See figure 23 and requirement 5
24	Packing, class A, durometer 55 ± 5	1					Rubber, ZZ-R-768	See figure 14

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/10- 3/					Material/specification	Remarks
		001	002	003	004	005		
25	Packing, class A, durometer 55 ± 5		1				Rubber, ZZ-R-768	See figure 17
26	Packing, class A, durometer 55 ± 5			1			Rubber, ZZ-R-768	See figure 12
27	Packing, class A, durometer 55 ± 5				1		Rubber, ZZ-R-768	See figure 21
28	Packing, class A, durometer 55 ± 5					1	Rubber, ZZ-R-768	See figure 24
29	Ring, retaining	1	1	1	1	1	SPIROLOX RR-165-S, or equal	See figure 1
30	Ring, retaining	1	1	1	1	1	TRUARC 5100-225-W, or equal	See figure 1
31	Nut	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 9
32	Nut, gland	1	1	1	1	1	Ni-Al-BRZ, QQ-C-465	See figure 4 and requirement 2
33	Washer, 4-3/8 od by 2.358 id by 0.162 thick	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 1
34	Pin, positioning	2	2	2	2	2	M24231/5-054	See figures 10, 15, 18, 19, and 22
35	Pin, guide	1	1	1	1	1	M24231/5-071	See requirements 4 and 6
36	Insert	1	1	1	1	1	M24231/5-077	See figure 1

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/10- 3/ 001	002	003	004	005	Material/specification	Remarks
37	Screw	1	1	1	1	1	M24231/5-083	See figure 1
38	Cap, pressure-proof	1	1	1	1	1	M24231/14-011	See figure 1
39	Wire, 15 feet	As required					Type B-16, MIL-W-16878/1	See figure 1
40	Pin, contact	14	24	30	40	40	MIL-C-24231	See figure 3
50	Cable, type 2SWU-1, 15 ft						MIL-C-24643/33	See figure 25
51	Assembly, retainer plate	1					Ni-Cu, class A, QQ-N-281	See figure 15 and requirement 5
52	Assembly, retainer plate		1				Ni-Cu, class A, QQ-N-281	See figure 18 and requirement 5
53	Assembly, retainer plate			1			Ni-Cu, class A, QQ-N-281	See figure 10 and requirement 5
54	Assembly, retainer plate				1		Ni-Cu, class A, QQ-N-281	See figure 19 and requirement 5
55	Assembly, retainer plate					1	Ni-Cu, class A, QQ-N-281	See figure 22 and requirement 5
56	Epoxy, resin, molded	As required					Cast epoxy resin, compound 4/	See figure 3

1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.
4/ Arcoset No. 200, M.M. Reif and Co., Rockledge, PA, or Phillicast No. 2, Philadelphia Resin Corp., Montgomeryville, PA, or equal.

NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

3. The flange outside diameter is intended to coincide with commonly available 3-1/2 inch bar stock. The dimension may vary after machine clean-up.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-10)

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